CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

Fundamentals of Database Management Systems (ITUC202)

Practical File – ODD Semester (2026-27)

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Practical 1 : Understanding the Data Engineering Lifecycle

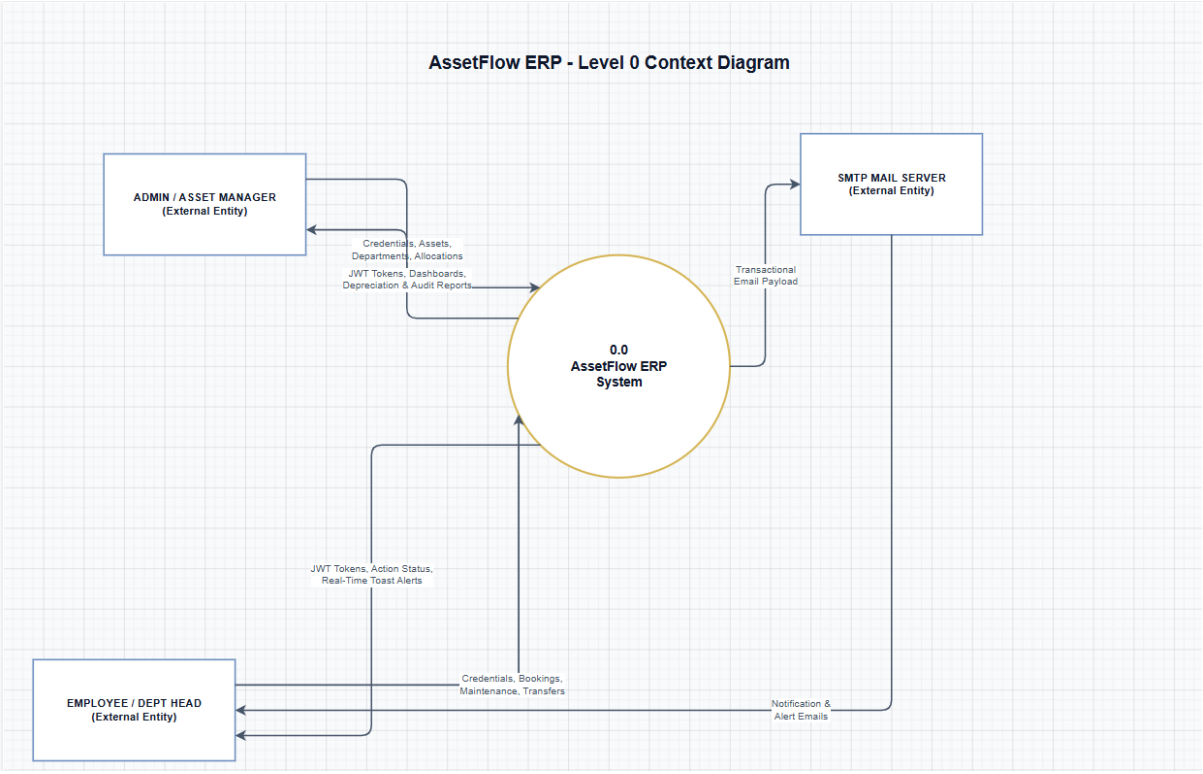
Problem Definition: Study, dissect, and map the complete end-to-end data engineering

lifecycle for an enterprise e-commerce platform. Students must trace data states across all five core lifecycle stages: data generation, ingestion, storage, transformation, and serving. The solution requires creating a detailed architectural data flow diagram along with a system manifesto documenting cross-cutting lifecycle undercurrents (security, observability, and data privacy). This initial structure must be committed to a personal Git repository.

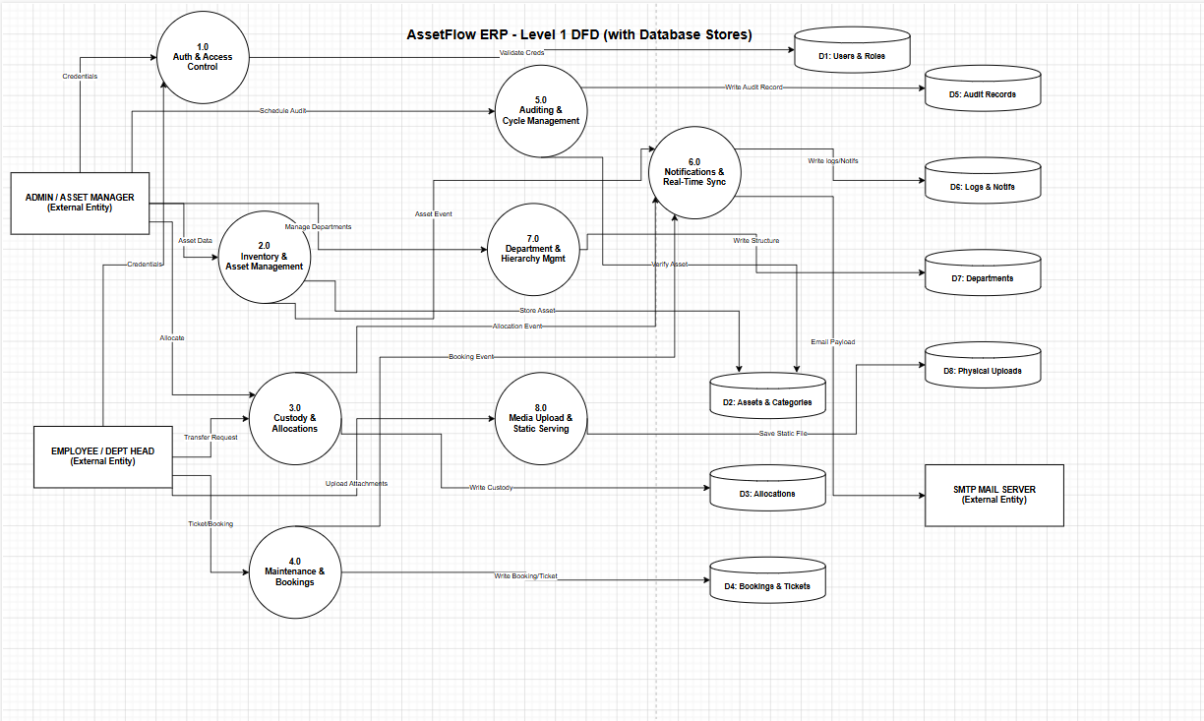
Dataset/Test Data: Mock enterprise e-commerce dataset schema definition (containing transactional database entities, clickstream tracking JSON payloads, and third-party inventory log streams).

Solution :

- DFD level 0



- DFD level 1



Key Questions / Analysis / Interpretation to be evaluated during/after

Implementation:

● **Que / Key Point 1:** How do the technical demands of downstream serving layers directly influence ingestion and storage design decisions during the early generation phases?

Ans : The serving layer needs fast and accurate data for dashboards, reports, and analytics. Therefore, data is collected in the correct format during generation and ingestion. Storage is also designed to organize data efficiently so it can be retrieved quickly when needed.

● **Que / Key Point 2:** Differentiate between the broader data lifecycle and the technical data engineering lifecycle within an enterprise.

Ans : The data lifecycle covers the complete journey of data from creation to deletion. The data engineering lifecycle focuses on the technical processes like data ingestion, storage, transformation, and serving. It ensures that data is reliable, scalable, and ready for analysis.

● **Que / Key Point 3:** Identify how the undercurrents of security and data observability should be enforced at each lifecycle boundary.

Ans : Security is maintained using authentication, authorization, and data encryption at every stage. Observability is achieved through logging, monitoring, and alerts to track data flow and detect issues. These practices help keep the data pipeline secure, reliable, and efficient.